

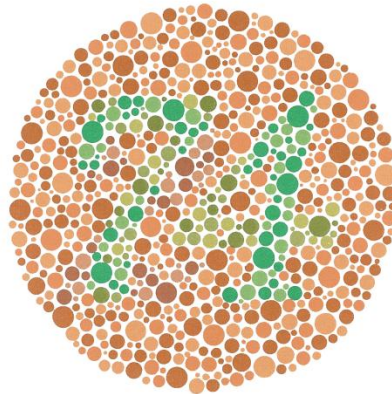
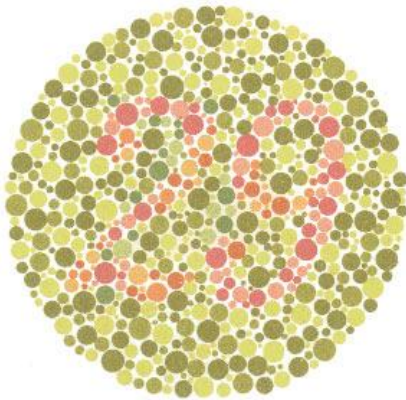
# Sight

- Visual system workings
- Color – color blindness : 8% males, 1% females
- Much done by context & grouping (words optical illusions)

# Visual perception

Red – green color blindness (daltonism) is very common (8% of adult males)

Don't use only color to highlight differences, or use choose colors that are difficult to confuse



# Visual perception

Colors of different wavelengths are hard to tell apart

Don't use red text on blue background



with age, blue becomes harder to read

## Change Blindness

difficult to see visual changes when our vision is interrupted Avoid abrupt changes in the interface (show animations, highlight changes)

# Visual Perception

- Vision involves constant movement – of our eyes, head and body, or the things we see or all of the above
- The outer parts of our retina are particularly sensitive to motion, so when something comes into our field of vision, our attention is drawn to it.
- Even the fact that we have two eyes (binocular vision) is a kind of movement: the two views are slightly different, as if we had moved a few inches to the left or right.

# Visual Perception

- Retina is light sensitive and contains 2 types of photoreceptors: rods and cones. Rods are highly sensitive and allow vision of weakly illuminated objects. They dominate peripheral vision
- Cones are grouped in the center and since there are 3 types, sensitive to different wavelength of light they allow color vision

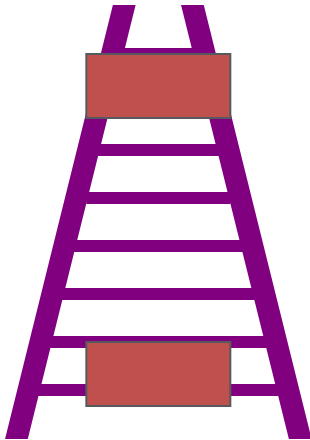
# Visual Perception

- Vision is the main source of information : 75% of information perceived by human is of visual nature
- The fastest signals reach the cortex in about 30 ms. Color perception needs about 80 ms to reach: this is faster than movement perception

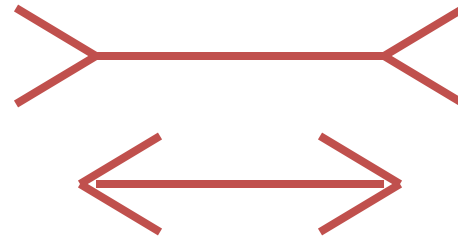
# Visual perception



# Optical Illusions



the Ponzo illusion

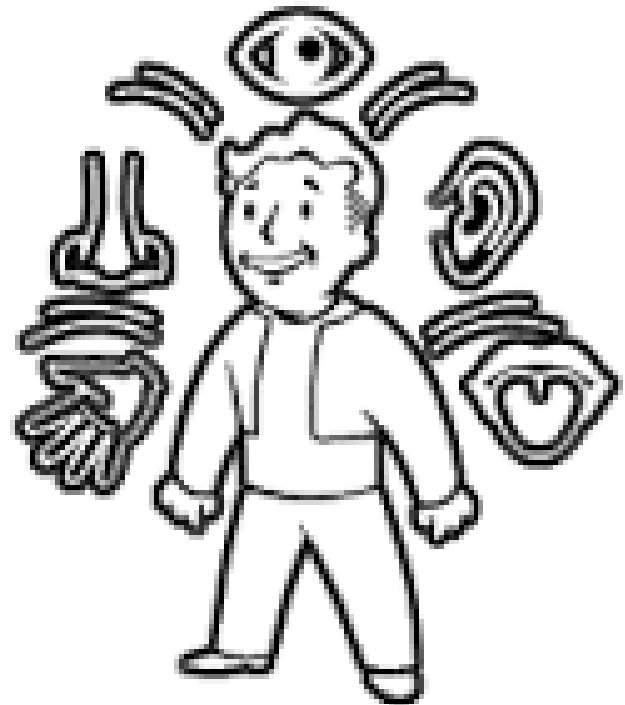


the Muller Lyer illusion



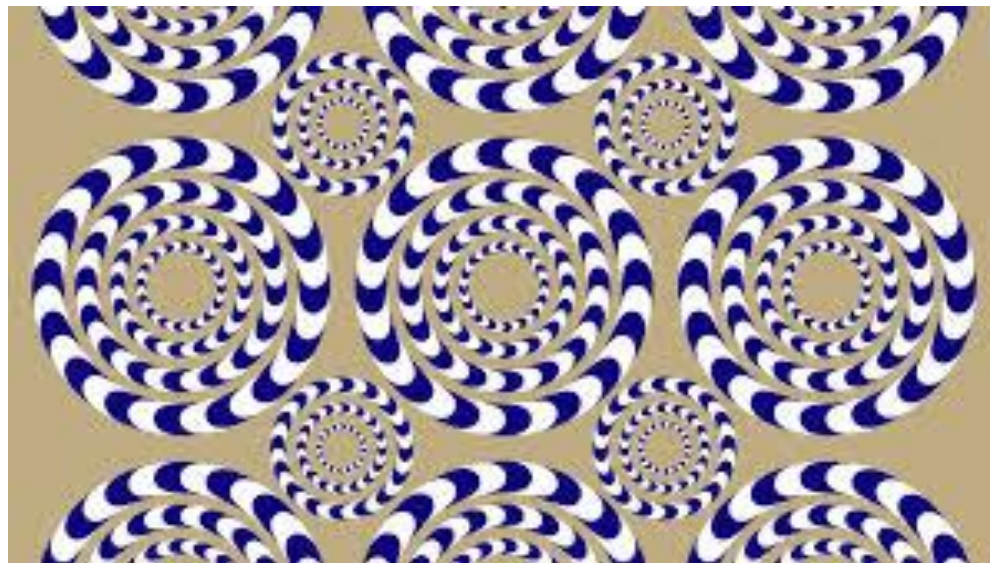
# Gestaltism and law of perception

- Gestalt is a psychology term which means 'unified whole'. It refers to theories of visual perception developed by German psychologists in the 1920s



# Gestaltism and law of perception

- These theories attempt to describe how people tend to organize visual elements into groups or unified wholes when certain principles are applied.



# Gestaltism and law of perception

- If we see what is not there, what is it what we are seeing? You could call it an illusion, but it's not an hallucination
- Wetheimer explained that you are seeing an affect of the whole event, not contained in the sum of the parts



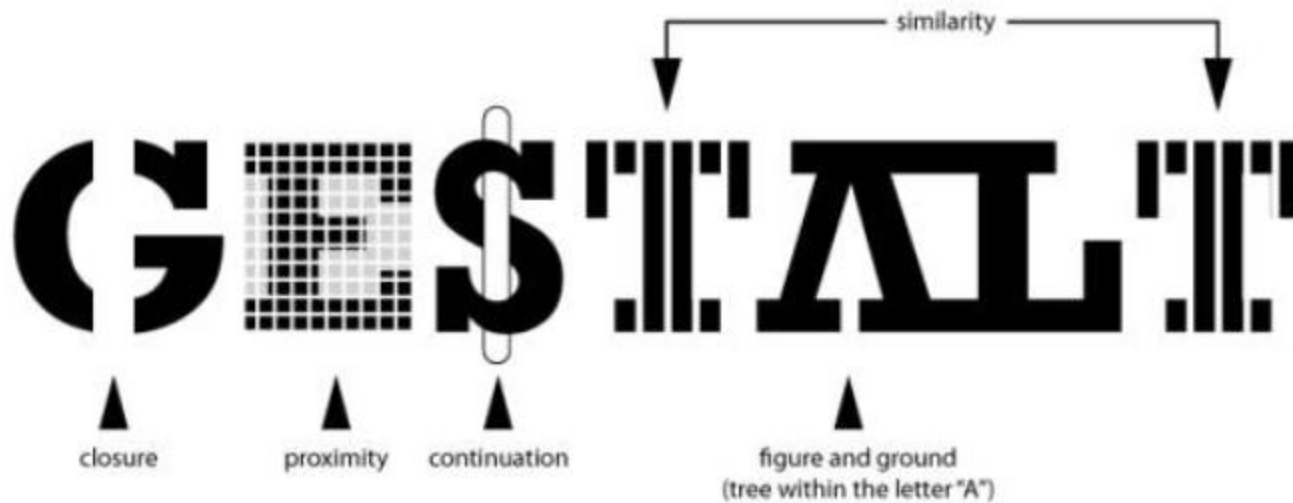
# Gestaltism and law of perception

- We see a coursing string of lights, even though only one light lights at a time, because the whole event contains relationships among the individual lights that we experience as well.





- Get into a group





- Get into a group and define following term:
  - A. Continuity
  - B. Proximity
  - C. Similarity
  - D. Closure
  - E. Common fate
  - F. Past experience
  - G. Figure-ground

# Gestalt laws of perception

- Continuity
- Proximity
- Similarity
- Closure
- Common fate
- Past experience
- Figure-ground

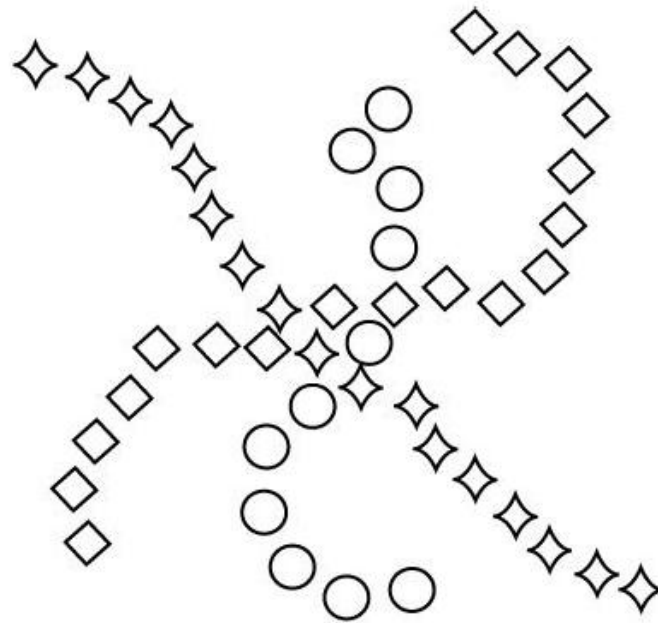
these laws act at the same time and can be occasionally contradictory

# Gestalt Principles



# Gestalt Principles

- **Continuity** – Continuation occurs when the human eyes follow the direction from one object to another, perceiving separate objects as one



<< Continuity >>

# Gestalt Principles

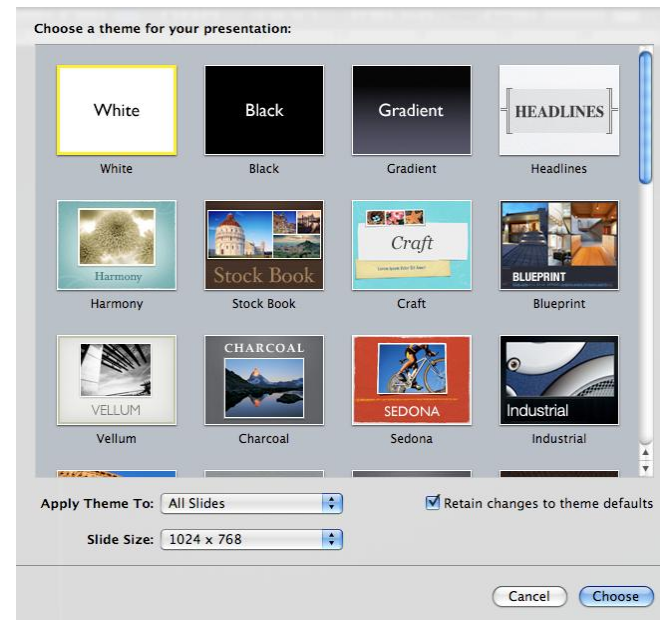
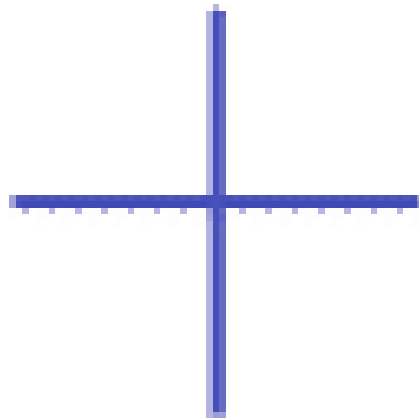
- Continuity in the form of a line, an edge, or a direction from one form to another creates a fluid connection among compositional parts
- The smooth flowing crossbar of the 'H' leads the eye



# Continuity

we tend to perceive elements grouped together, and integrated into perceptual <<wholes>> if they are aligned

e.g. different style options in a UI presented one after another



# Gestalt Principles

- **Proximity** – Objects placed close together are often perceived as a group



# Gestalt Principles

- Also called 'grouping', the principle concerns the effect generated when the collective presence of the set of elements becomes more meaningful than their presence as separate elements



# Gestalt Principles

Grouping of this sort can be achieved with: Tone/value

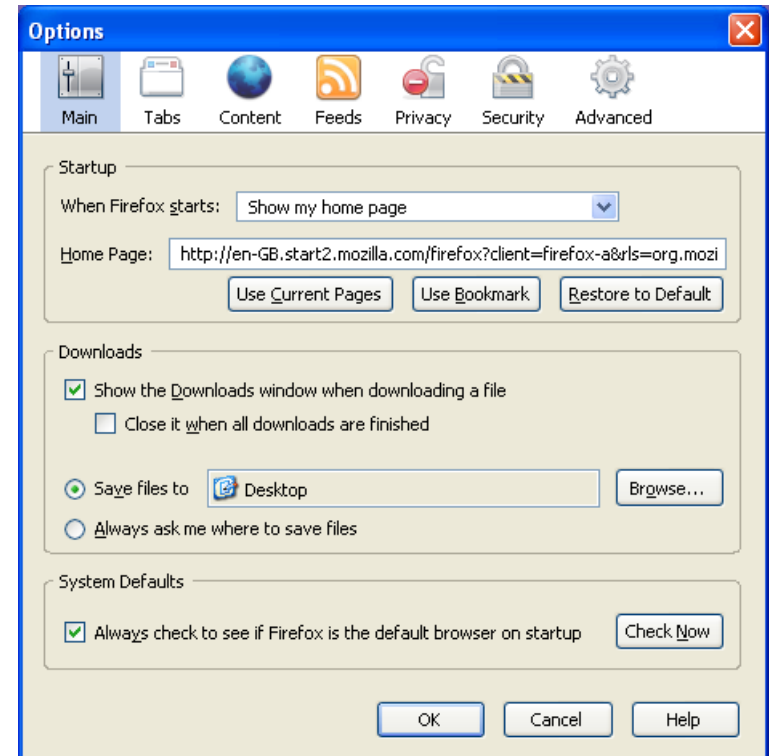
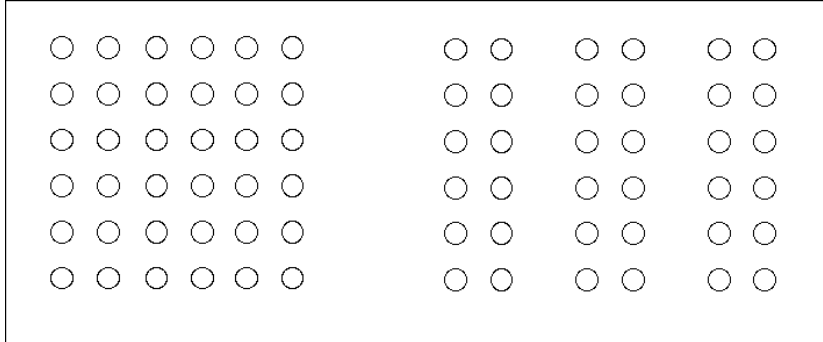
- Color
- Shape
- Size
- Or other physical attributes



# Proximity

we group objects first by their proximity between them

e.g. functions in a dialogue box



# Gestalt Principles

- **Similarity** – Similar objects are often perceived as a group





# Gestalt Principles

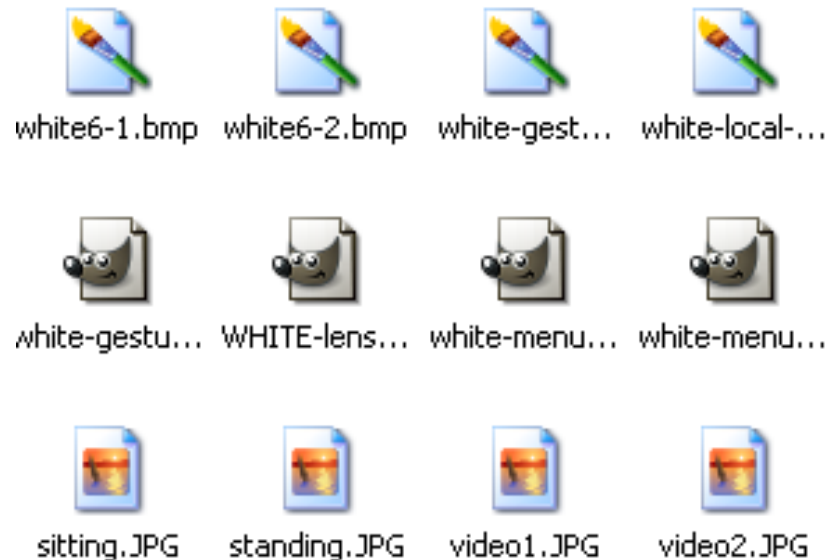
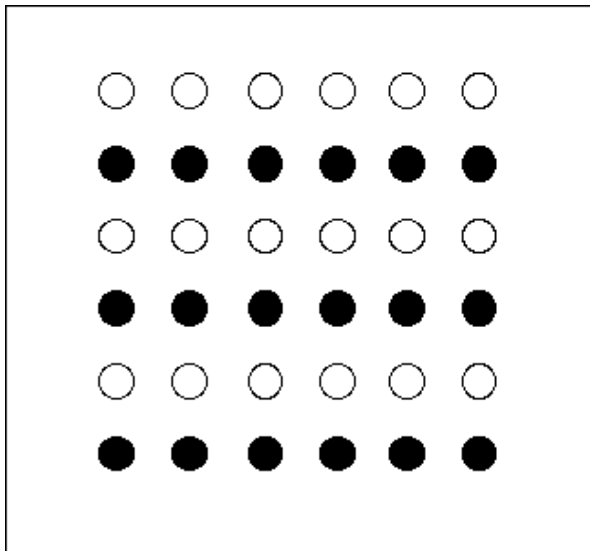
- When similarity occurs, an object can be emphasized if it is dissimilar to the others
- This is called anomaly



# Similarity

if distance (proximity) does not allow grouping, we tend to group objects base on their perceived similarity in form

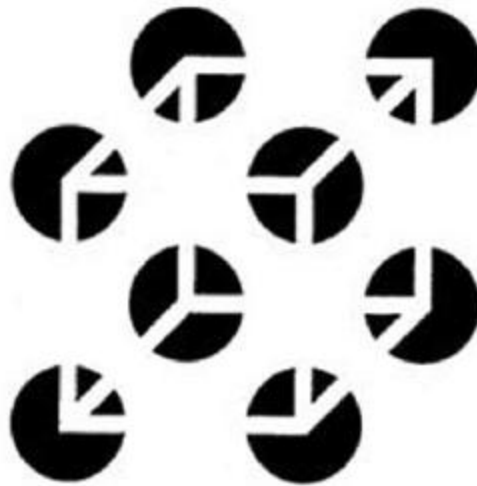
e.g. similar file icons to visually organize and remember their application (shape, size, color)



# Gestalt Principles

- **Closure** – When gaps appear between shapes, people tend to mentally close those gaps and form a perception of a whole object

**G E S T A L T**



# Gestalt Principles

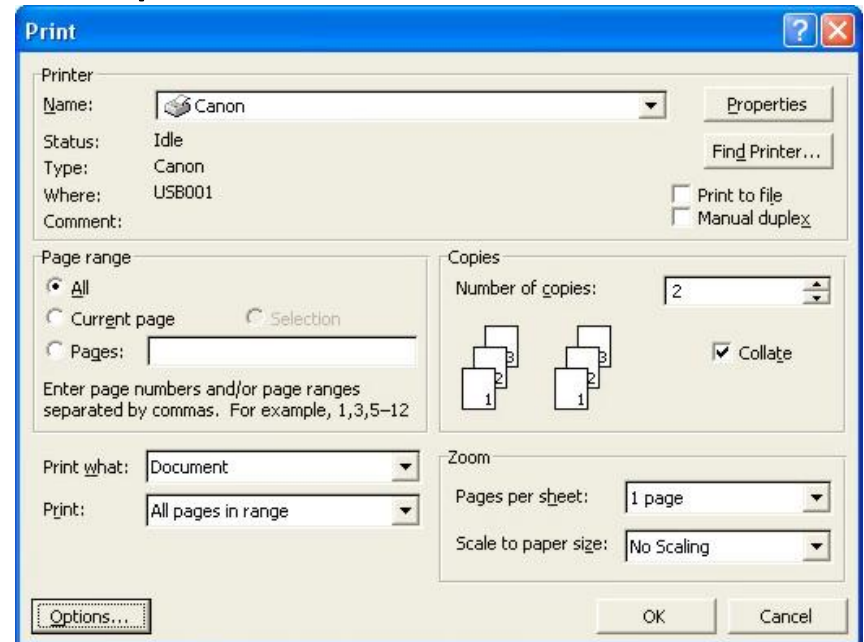
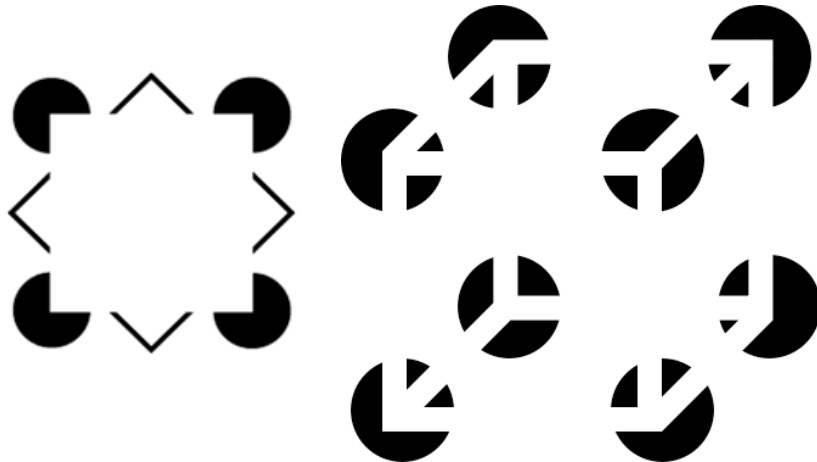
- Imaginary lines called vectors, or shapes called counter forms, are generated by these relationships, which the eye understands as part of the composition even though there is nothing there.



# Closure

we perceive objects such as shapes, letters, pictures, etc., as being whole even when they are not complete (we complete the missing parts)

e.g. we can group by explicit or implicit borders items in a UI



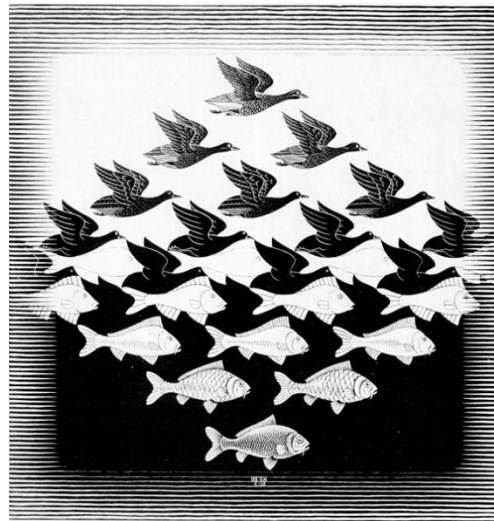
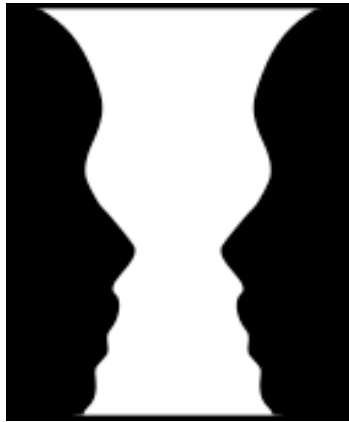
# Gestalt Principles

- **Figure and Ground** – Differentiate an object from its surrounding area that are formed by the foreground (figure) and background (ground)



# Figure - ground

Perception consists of a distinction between the graphical figure (target) and ground (context).  
It should always be clear in the UI

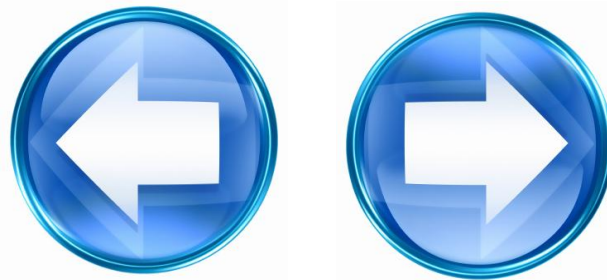


# Symmetry

symmetries are aesthetically pleasing, we tend to group symmetrical objects as one group with a central point

e.g. symmetrical in the UI have symmetrical icons and are seen as a group

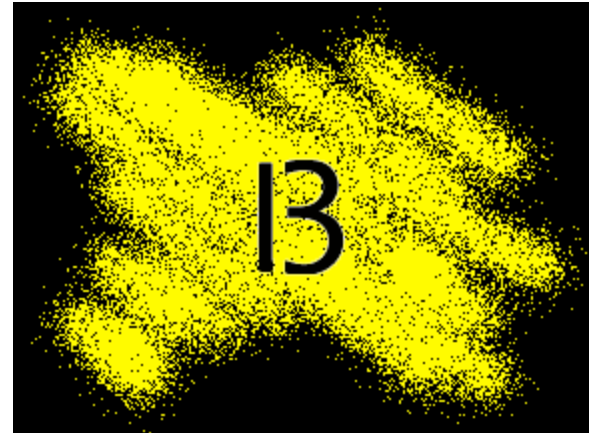
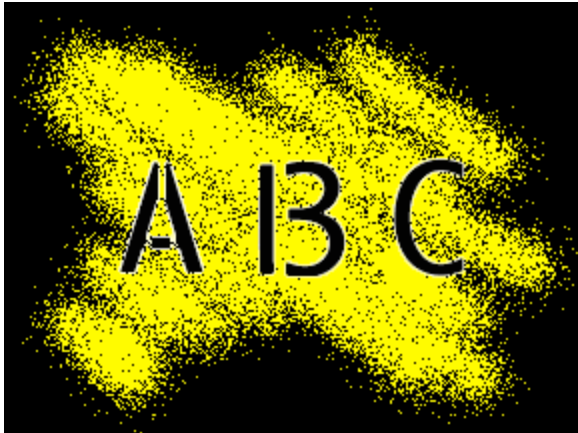
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# Past experience

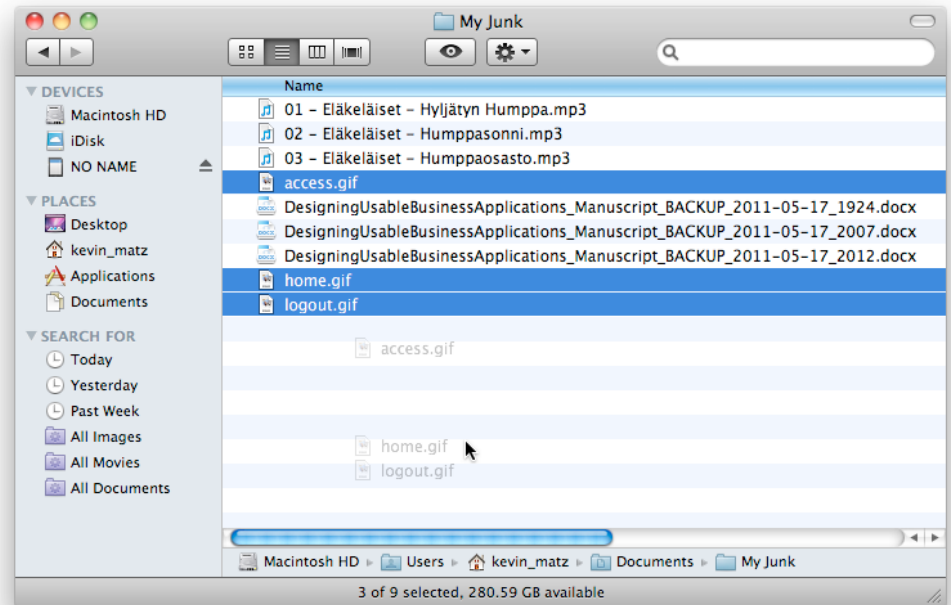
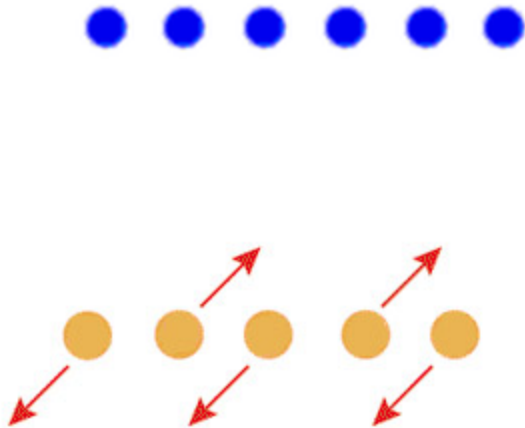
past experience and context affect the interpretation of elements in a group



# Common fate

elements moving in the same trajectory with the same speed  
are seen as a group

e.g. if you select and drag some icons, shadow of these items all  
move at the same direction and speed



# Perception

## Sensory organs : ear

Frequencies : 20 Hz to 20 KHz

Selective focus and perception

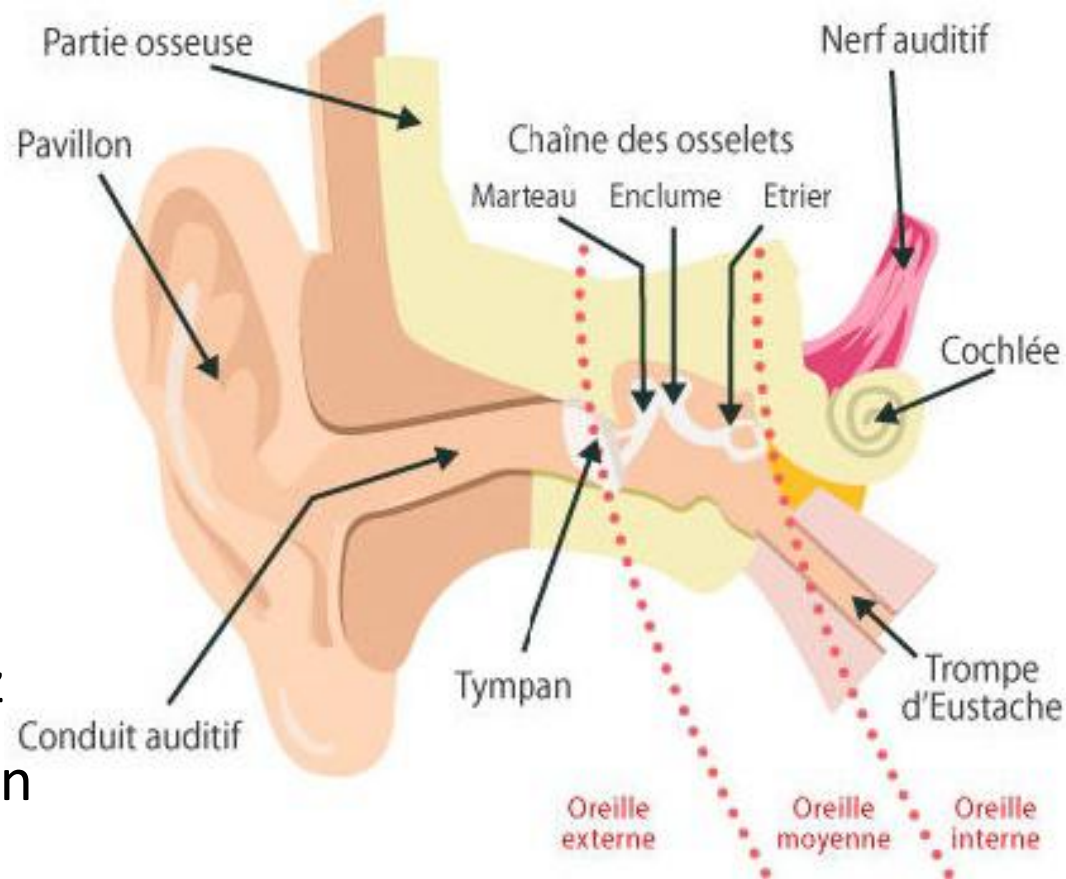
Masking Effects

we hear one sound source but not another

many factors affect it: frequency interval,  
intensity, distance

Localization of a source

correlation with visual localization



# Hearing

- Often taken for granted how good it is
  - Pitch – frequency
  - Loudness – amplitude
  - Timbre – type of sound (instrument)
- Sensitive to range 20Hz – 22000Hz
- Limited spatially, good temporal
- Performance

# Hearing

- Sounds can be perceived as coming from a location
  - Not terribly accurate
  - Cone of confusion

# Auditory perception

Listening needs less effort than reading

Reading is faster than listening

Written language is more permanent than audio

audio menus should be brief

Individual differences and limitations

individual preferences and capabilities

(reading) difficult for dyslexics

(listening) easy to forget

- multimodal interfaces

# Auditory perception

- Auditory perception is the ability to perceive and understand sounds, usually with specific organs, such as a human's ears
- Sound exists in the form of vibrations that travel through the air or through other substances
- Ears detect such vibrations and convert them into nerve impulses, which are sent to the brain where they can be interpreted.

# Auditory perception

- Deafness describes a condition in which individuals have no auditory perception; deaf individuals are not capable of perceiving or interpreting sounds
- Different animals can perceive different sounds; dogs, for example, are capable of perceiving very high-pitched sounds that humans cannot perceive



# Auditory perception

- There are many factors that affect auditory perception beyond simply hearing sounds. The brain is largely responsible for many processes that can turn a mass of incoming noise into something useful and understandable
- Auditory discrimination is the process by which one is able to note the differences between sounds; this is extremely important to language as spoken words are understood based on different sounds

# Auditory perception

- Discrimination between foreground and background is also an important part of auditory discrimination
- It is important to be able to focus on important noises and to ignore irrelevant and unimportant noises so that one is not overwhelmed by a vast amount of noise

# Auditory perception

- Auditory synthesis is another process very important to the comprehension of language
- It describes the process by which the brain combines different sounds into understandable units, similar to the way letters are combined into words and words into sentences

# Auditory perception

- Auditory sequencing is a process closely related to both memory and auditory perception
- It describes the ability to understand and remember the order in which certain sounds happened
- Individuals who have hearing problems may simply have difficulty hearing quiet sounds or extreme pitches.

# Auditory perception

- They may also, however, have problems with the above processes that are essential to making sense of sounds
- Problems with auditory perception can exist from birth, or they can be caused by injuries to the brain or ears
- They are generally easy to detect; one with hearing problems either cannot hear well or cannot make sense of the sounds that he hears

# Touch perception

Tactile sense: receptor types

temperature (hot, cold), hardness, pressure, touch

Sense that is also proprioceptive (vs. exteroceptive)

configuration of our own body in the space,

feeling as objects/tools are extensions of our own body

Kinesthesia

Perception of the effort of muscles

and so of resistance/weight of object



# Touch

- Three main sensations handled by different types of receptors
  - Pressure (normal)
  - Intense pressure (heat/pain)
  - Temperature (hot/cold)

# Touch perception

- Provides important feedback about environment
- May be key sense for someone who is visually impaired
- Stimulus received via receptors in the skin
  - Thermoreceptors
- Heat and cold
  - nociceptors
- Pain
  - mechanoreceptors



# Touch perception

- Pressure (some instant, some continuous)
- Some areas more sensitive than others eg. Fingers
- Kinethesis – awareness of body position
  - Affects comfort and performance
- Haptic perception is produced by the hand movements getting in contact with objects

# Touch perception

- This type of perception is of great importance when users have to operate on virtual realities: robotics, simulators, video games..
- This is also true for blind people, using the Braille alphabet
- Examples in robotics
  - Medical instrumentation used for remote surgery
  - Instrumentation for manipulating radioactive objects

# Processors

each processor has a processing cycle

Necessary duration to treat an input and produce an output

Speed of processing depends on individual humans and external conditions

(eg. Intensity of the stimulus, noise, alcohol,...)

the fastest speed can be 10 times faster than the slowest



# Perceptual fusion

Two stimuli in the same cycle ( $T_p \approx 100\text{ms}$ ) are combined

e.g.

two successive images that appear in the same cycle can be perceived as one event: animation.

a sound and a movement at the same time may be understood as the same event

*End of Lecture*

Previous Lecture

**MVC**

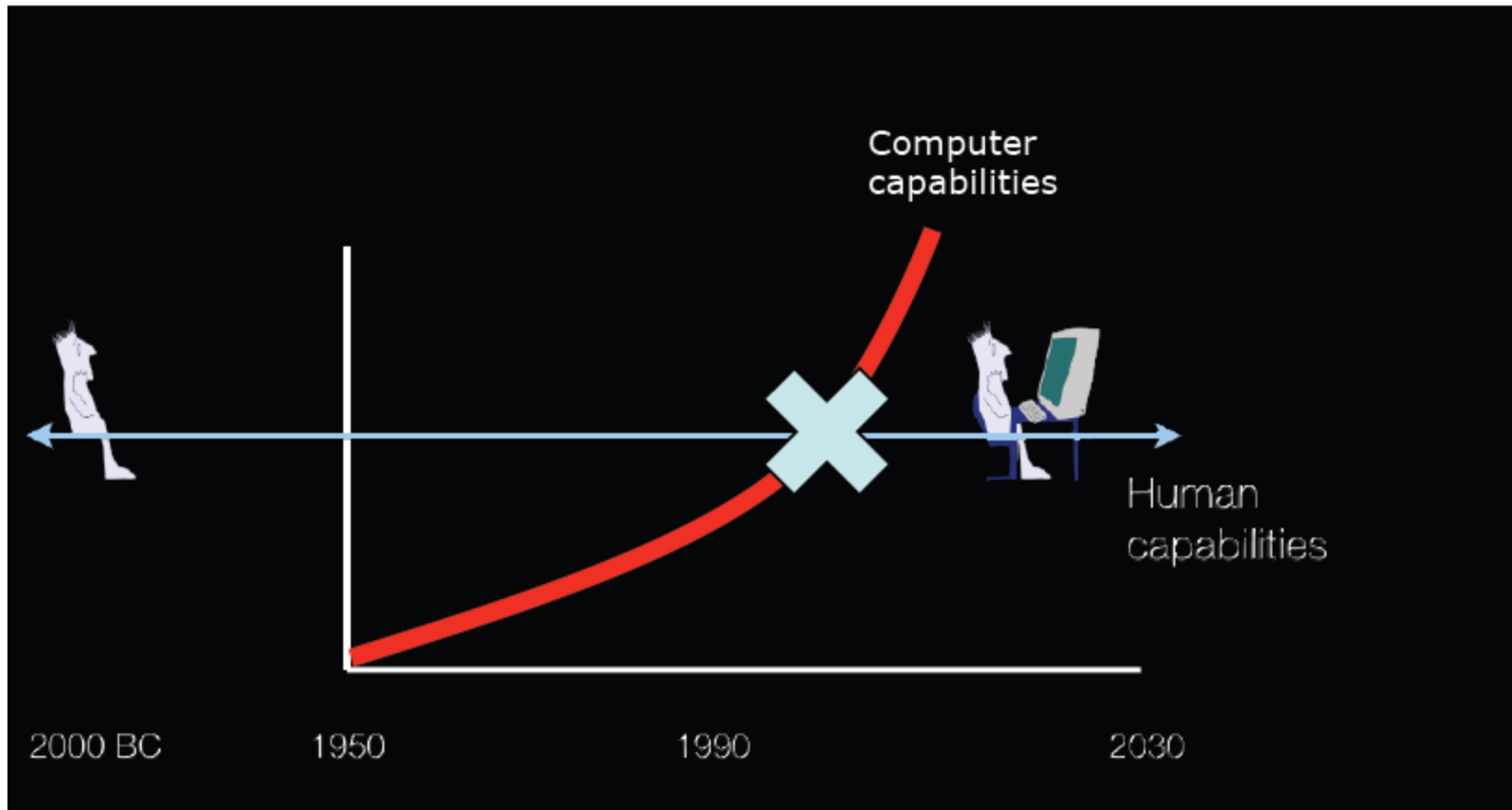
# PSYCHOLOGY AND MOTOR PERFORMANCE



## CHAPTER FIVE

Instructor Name :- Dr. Suzana Binti Ahmad

# Human Capabilities



Saul Greenberg



# Human / Stakeholder

Defined as anyone who is affected by the success and the failure of the system



# Categories of Stakeholder

## 1. Primary

- Those who use the system

## 2. Secondary

- Those who don't directly use the system but receive the output from it
- Someone who receive reports from the system



# Categories of Stakeholder

## 3. Tertiary

- Those who do not fall into 1 and 2 but who are affected by the success or failure of the system
- A business competitor whose profits increase or decrease depending on the success of the system



# Categories of Stakeholder

## 4. Facilitating

- Those who are involved in designing, developing and maintaining the system





## An **Airline** Reservation System

“An international airline is considering introducing a new booking system for use by associated travel agents to sell flights directly to the public”

Classify the stakeholders ..

1. Primary
2. Secondary
3. Tertiary
4. Facilitating



# Classifying stakeholders

- **Primary Stakeholders**

- Travel Agency staff, airline booking staff

- **Secondary Stakeholders**

- Customers, Airline management

- **Tertiary Stakeholders**

- Competitors, Civil aviation authorities, Airline Shareholders

- **Facilitating Stakeholders**

- Design team, IT Department staff



# Primary stakeholder

=

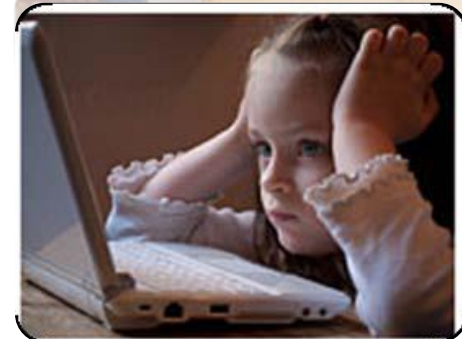
# USER



# Understanding Users

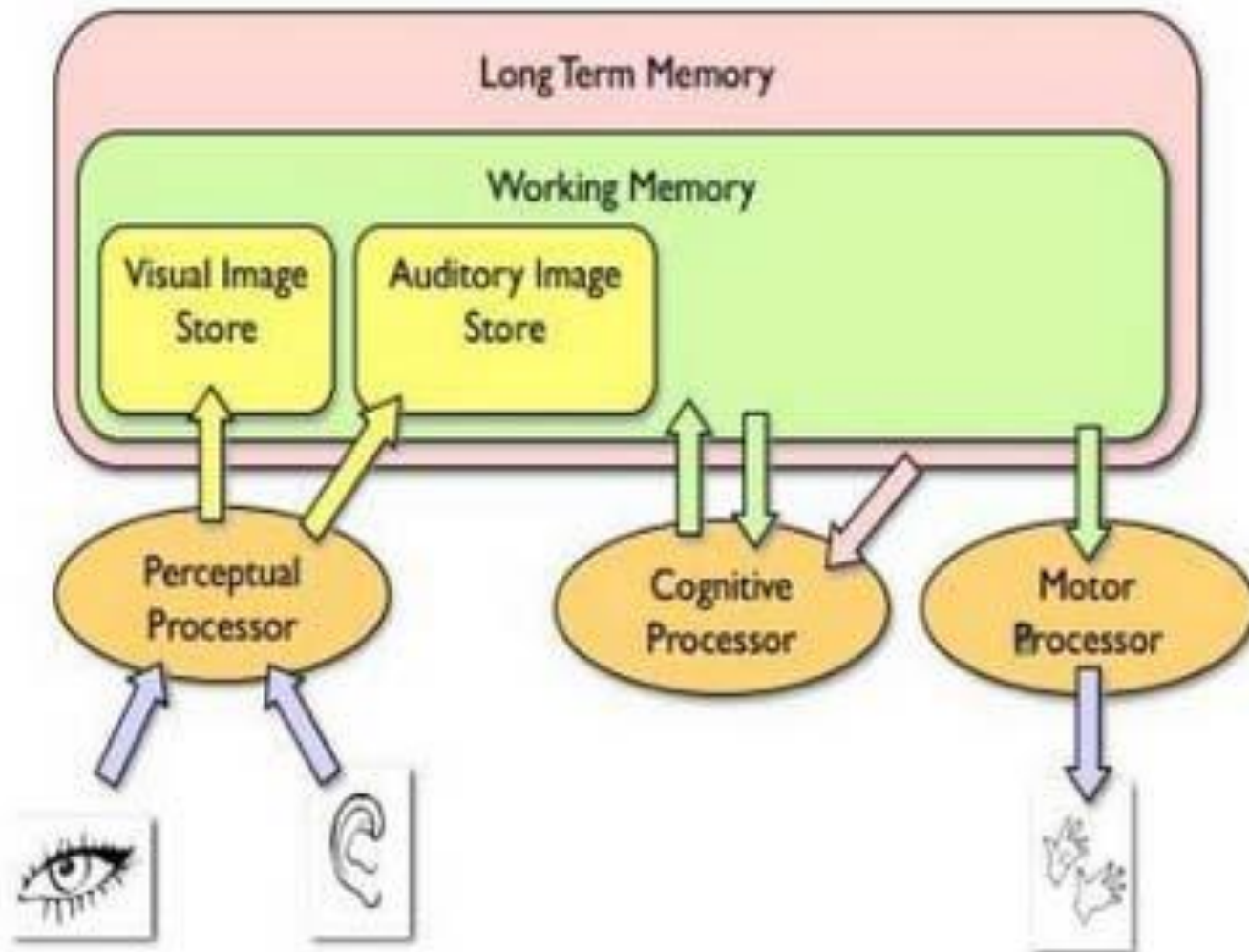
## Who are they?

- Individual user
- A group of users working together or a sequence of user in organization each dealing with some part of the task or process.
- Limited in their capacity to process information
- In order to design, it is important to understand the capabilities and limitations of those we are designing for.



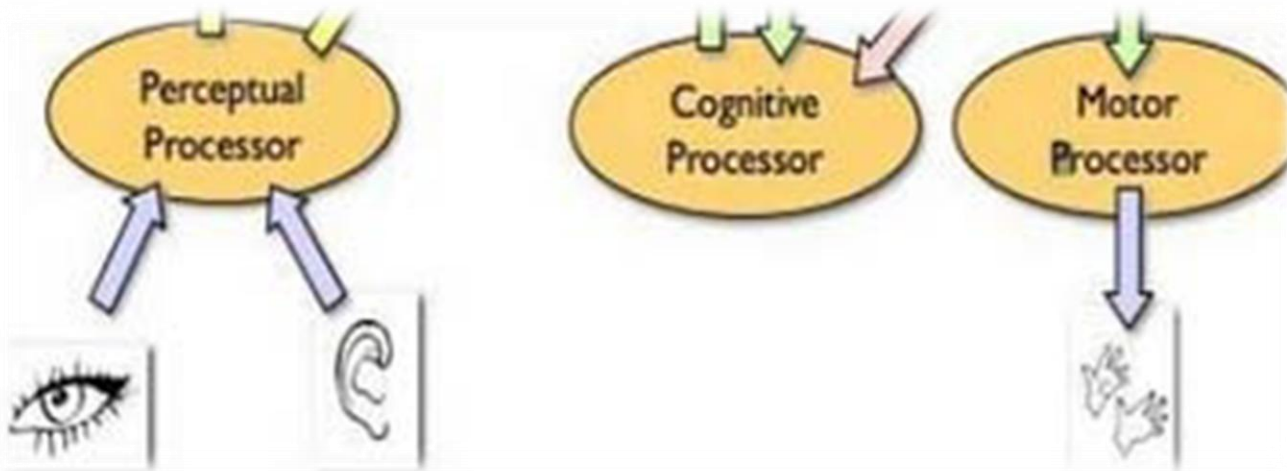


# Understanding Users



# Human capabilities

- Three interacting processor/subsystems:



- Perceptual (eg. Read – scan/hear )
- Cognitive (eg. Think)
- Motor (eg. Respond)

# Human capabilities

- Human computer interaction depends on what humans are actually capable of observing and articulating
- What do we know about human capabilities that could or should constrain interface design?
- Limits on perceptual capability – eg. Contrast, resolution
- Limits on motor capability – eg. Reach, speed, precision
- Limits on attention capacity

# Human capabilities

- Limits on memory
- Rates of learning and forgetting
- Causes of error
- Mental models & biases
- Individual differences (the average size fits few people)
- Variable state (eg. Stress, fatigue)
- Special needs & age

# Human

Want to improve user performance

- Know the user
  - Senses
  - Information processing systems
    - Perceptual
    - Cognitive
      - memory

# Senses

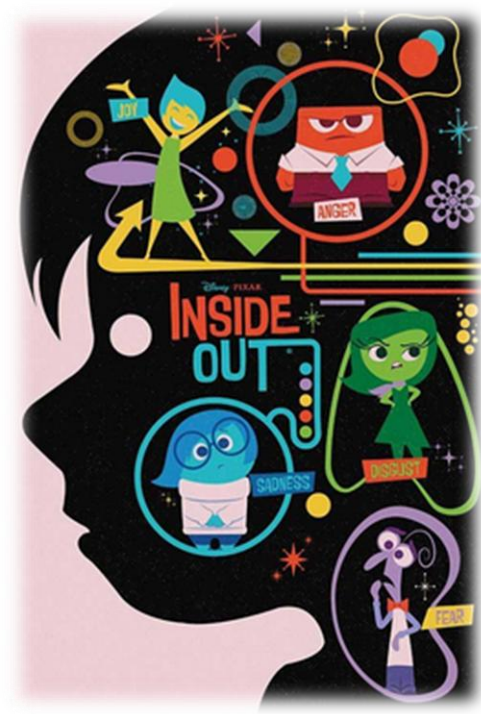
- Sight, hearing, touch – important for current HCI
- <https://www.youtube.com/watch?v=HvuoCNv9ufk>



# MEMORY

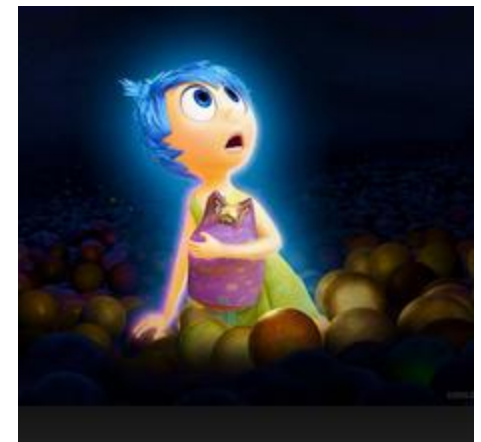


Long Term Memory



Short Term Memory

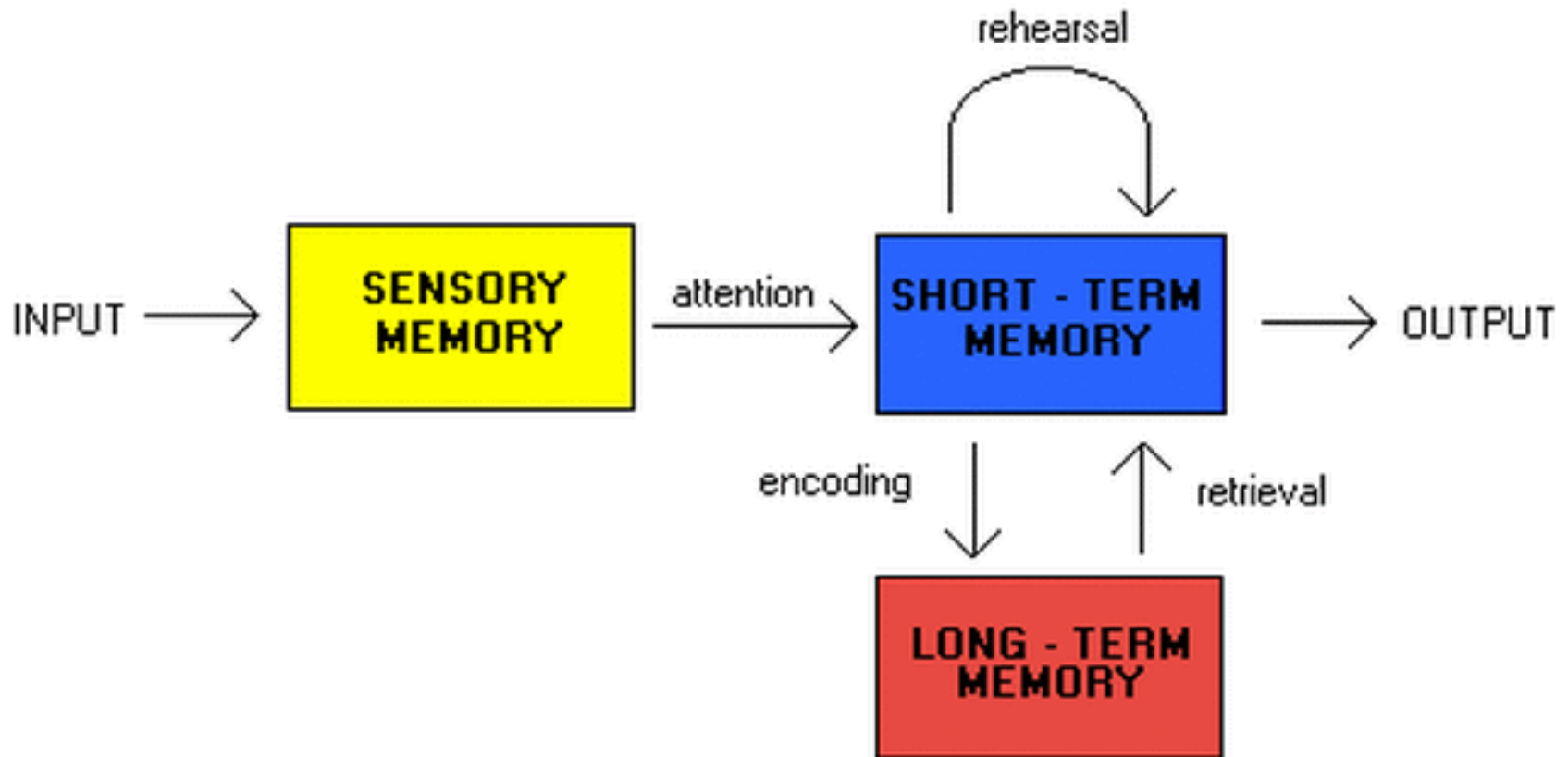
Working  
Memory



Memory Lost

# Sensory storage

- The 'input buffer' of the senses





# Sensory storage



- Stores most recent input unrecognized
- Storage time and capacity varies by type
  - Visual

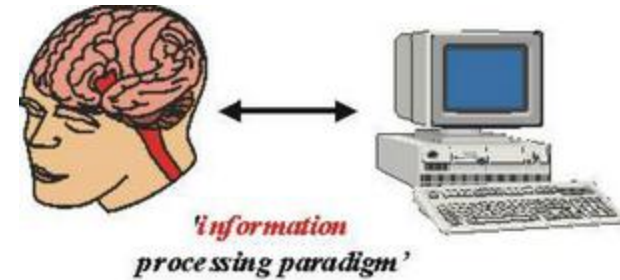
|                | Nominal         | Range            |
|----------------|-----------------|------------------|
| • Capacity :   | 17 text letters | [7 – 17] letters |
| • Decay time : | 200 ms          | [70 – 1000] ms   |
  - Audio

|                |                |                     |
|----------------|----------------|---------------------|
| • Capacity :   | 5 text letters | [4.4 – 6.6] letters |
| • Decay time : | 1500 ms        | [900 – 3500] ms     |

# Information processing systems

- Information i/o ...
  - visual, auditory, haptic, movement
- Information stored in memory
  - sensory, short-term, long-term
- Information processed and applied
  - reasoning, problem solving, skill, error
- Emotion influences human capabilities
- Each person is different -

**HUMAN FACTORS**



# Human Factors



We look at the human factors that affect how people interact with computers and computer programs:

- **Physiology** - physical make-up, capabilities
- **Cognition** - thinking, reasoning, problem-solving, memory
- **Perception** - how a person perceives what input they get through their senses

# Human Factors : Physiology

- The design of devices are often be affected by the human physiology
- Some **constraints** can be introduced and applied based on the physical built up of the users

## Examples:

- Keyboard keys cannot be smaller than finger size
- Smaller machines must use different input facilities
- Toilet for toddlers
- Specific door widths and heights (home, hospitals)
- Anymore?



# Human Factors : Physiology: Reaction Time



www.geetyimages.com (royalty free images)  
www.geetyimages.com (royalty free images)

Human reaction times:

- Audio signal - 150ms
- Visual signal - 200ms
- Pain - 700ms

**Examples of use:**

- Design of video games
- Traffic lights
- Phone

# Human Factors : Physiology: Movement

Speed or accuracy of movement are important for Interactive systems.



## Examples:

- Mouse - keyboard movement (affects choice of which devices/controls operate which actions of the system)
- Time taken to move to a target on screen
- Careful arrangement of menu items so that frequent choices are placed first

# Human Factors : Physiology: Disabilities

Designers must design so that disabled users can achieve maximum functionality and usability from computer systems.

## Examples:

- Speech input and output systems (useful for blind people and those with severe motor impairment)
- Keyboard pressing devices
- Eye movement detection devices





# Human Factors : Cognition

- The processes by which we become acquainted with things, how we gain knowledge, familiarity
- What goes in our heads when we carry out our everyday activities
- Involves understanding, thinking, remembering, reasoning, memorizing, attending, awareness, acquiring skills, creating new ideas.





# Human factors : Cognition

## – Managing Attention

- Process of selecting things to concentrate on at a point in time
- Depends on:

## 1. Users' goals

- ❖ If we know exactly what we want to find out, we try to match this with the information that is available

## Information presentation

- ❖ Greatly influence how easy or difficult



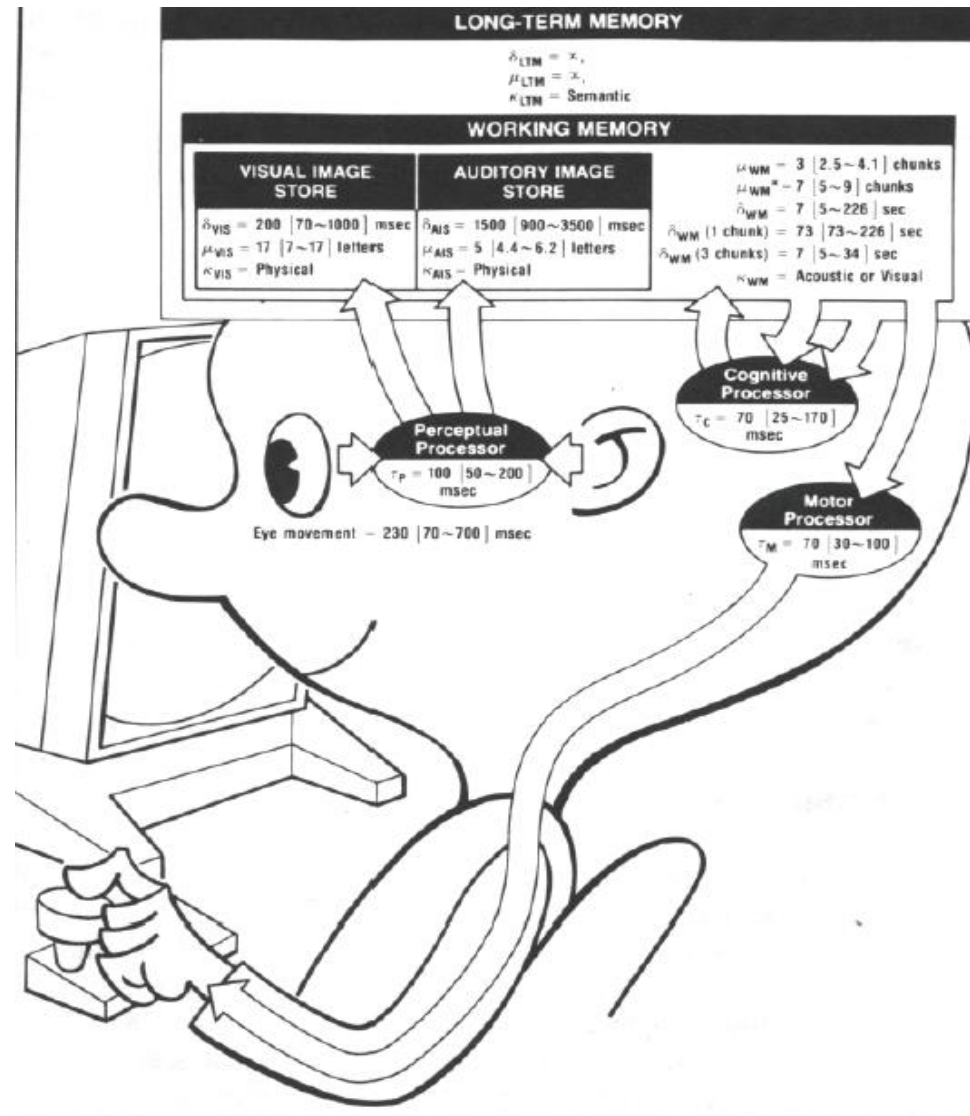
# Models of Human Performance

- Predictive
- Quantitative
  - Time to perform
  - Time to learn
  - Number and type of errors
  - Time to recover form errors
- Approximations

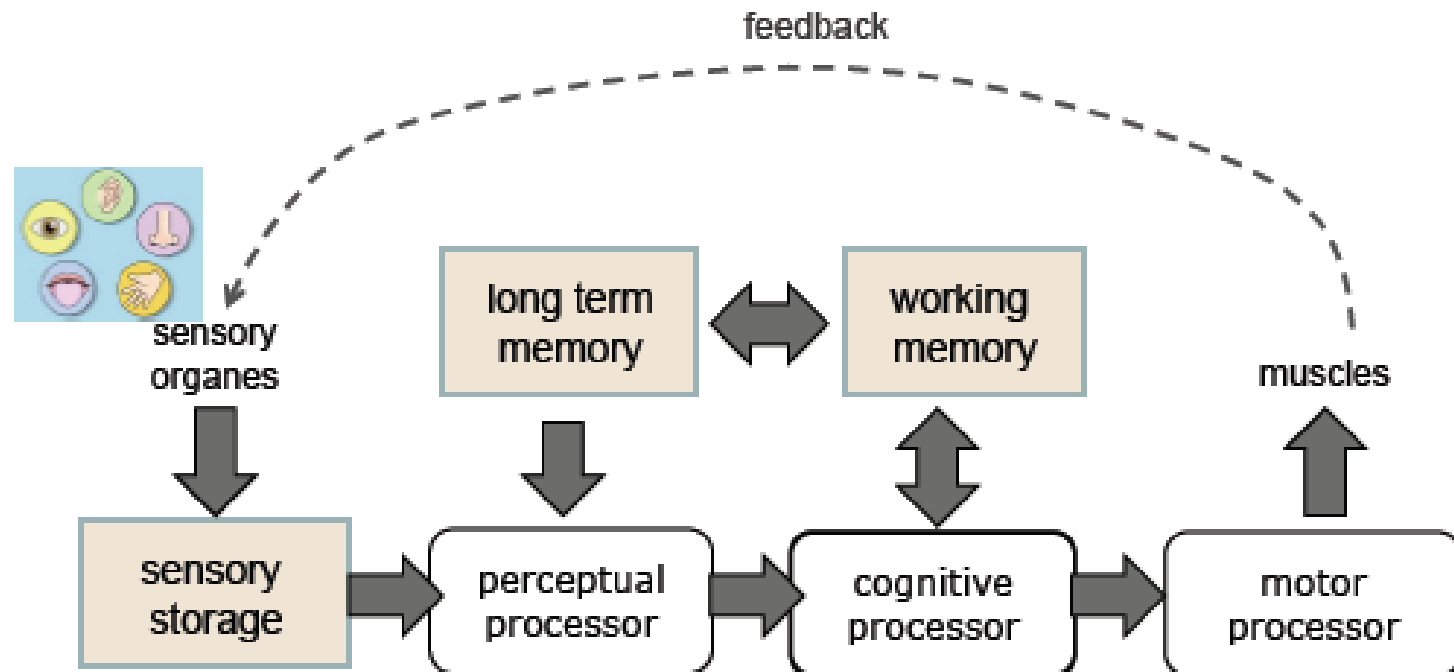


# Human Processor

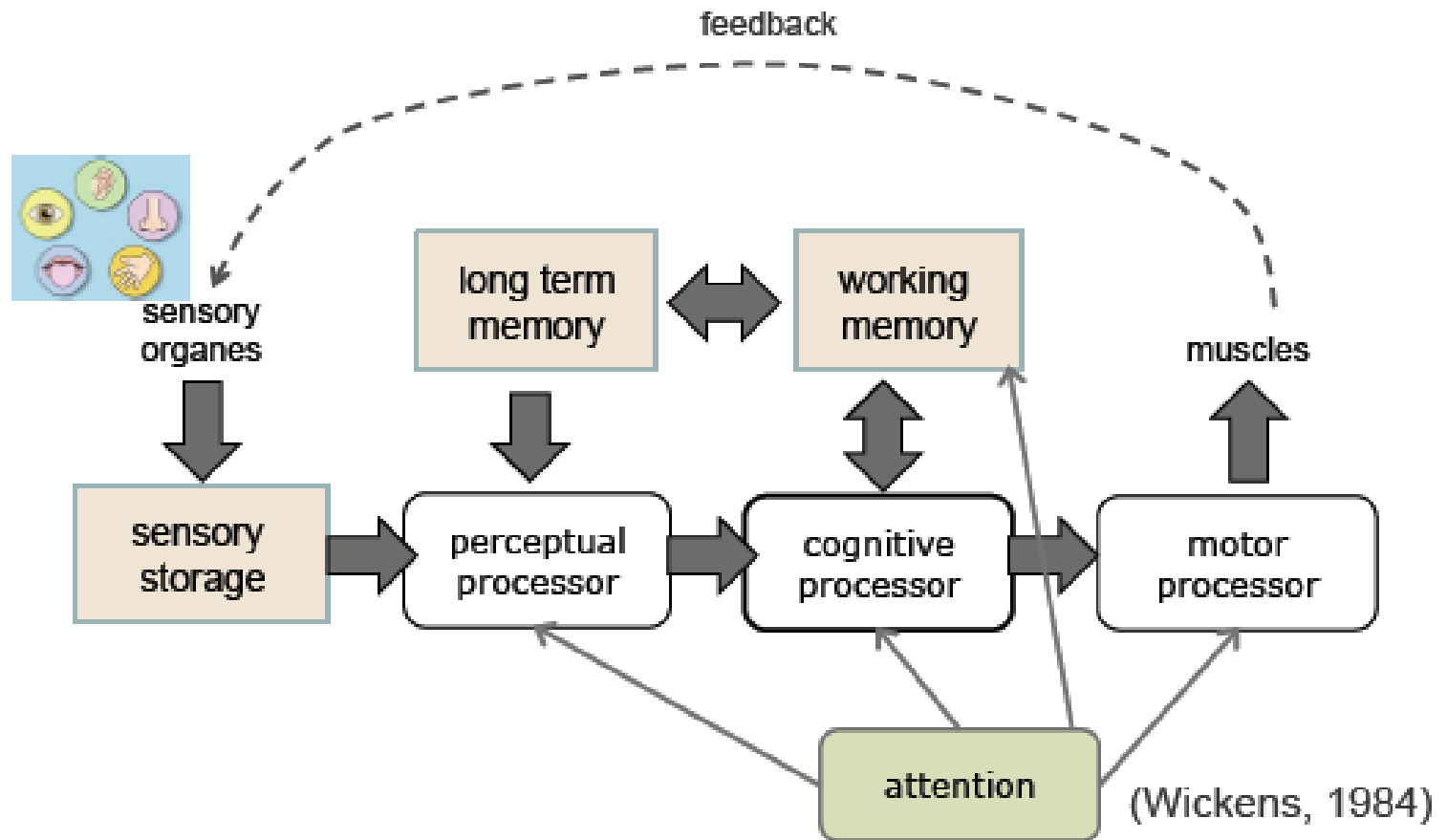
Modeling humans as  
an information  
processing system



# Human Processor



# Human Processor



# Processors



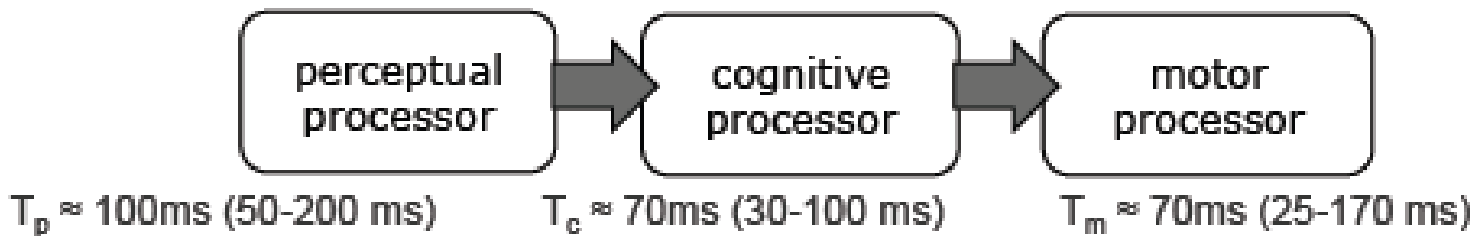
each processor has a processing cycle

Necessary duration to treat an input and produce an output

Speed of processing depends on individual humans and external conditions

(eg. Intensity of the stimulus, noise, alcohol,...)

the fastest speed can be 10 times faster than the slowest



# Human processor and perception

- Perception – seeing, hearing, touching, smelling, tasting, feeling the positions of joints and the tension of muscles, balance, temperature, pain..  
– begins with the stimulation of sensory neurons
- Each sense involves highly cells which are sensitive to a particular stimulus : pain receptors respond to certain chemicals produced when tissues are damaged

# Human processor and perception

- Touch receptors involve cells with hairs which, when bent, cause signals to travel down the cell's axon
- Balance, movement and even hearing involve similar cells
- Temperature sensitive neurons response to heat and cold



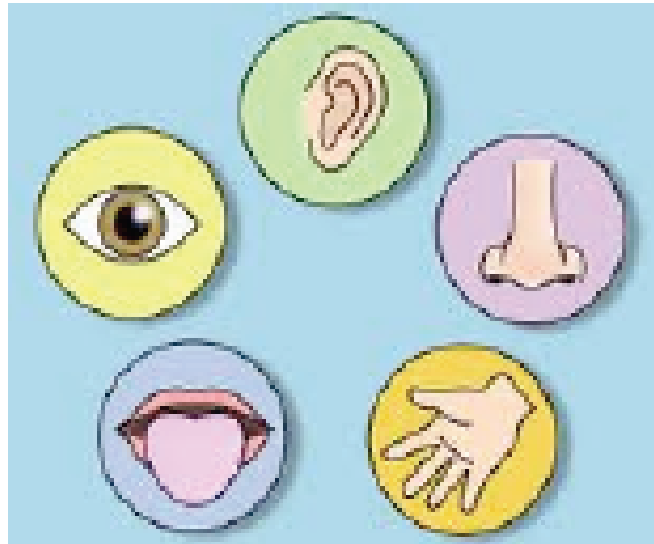
# Human processor and perception

- Taste and smell receptors respond to environmental molecules in the same way that other neurons respond to neurotransmitters
- And the neurons of the retina respond to the presence of light or the specific frequency ranges of light we perceive as color
- But perception is more than just passive reception of information

# Human processor and perception

- Perception is an active process: touch, for example, requires movement – something that nowadays we call ‘scanning’. Touch includes information about you (eg. Your muscles, joints) as well as about what you are touching
- We can say the same about hearing. We should really call it listening. The sound itself is intrinsically moving, of course – it is constantly changing. If it didn’t, we would stop hearing it

# Perception



# Perception

## Sensory organs : eye

Perception of color, movement, depth

Visual field 180° (x 160°)

Focus of attention

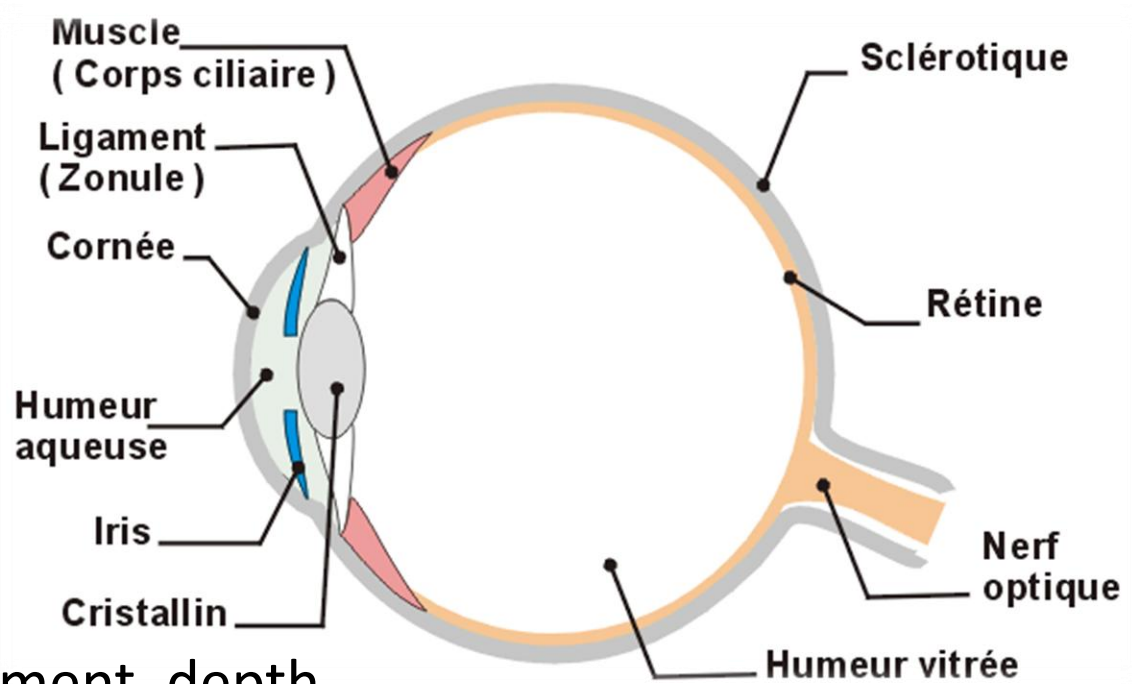
Visual acuity : 0,04 mm to 50cm (13° eye)

Fovea centralis (very high resolution) (3° eye)

## Peripheral Perception

Less sensitive to colors

More sensitive to movements



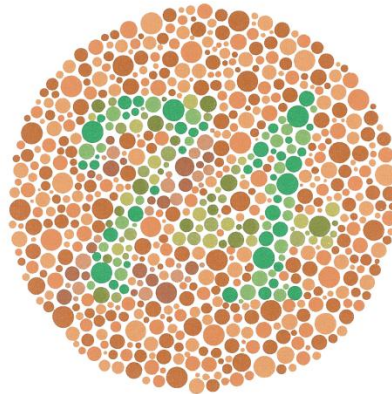
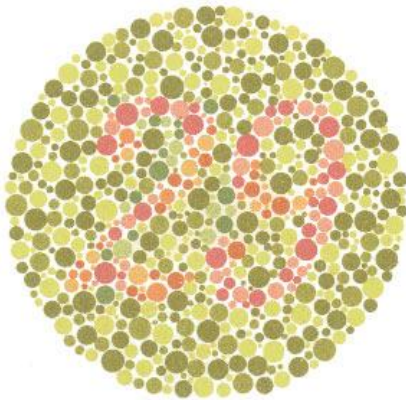
# Sight

- Visual system workings
- Color – color blindness : 8% males, 1% females
- Much done by context & grouping (words optical illusions)

# Visual perception

Red – green color blindness (daltonism) is very common (8% of adult males)

Don't use only color to highlight differences, or use choose colors that are difficult to confuse



# Visual perception

Colors of different wavelengths are hard to tell apart

Don't use red text on blue background



with age, blue becomes harder to read

## Change Blindness

difficult to see visual changes when our vision is interrupted Avoid abrupt changes in the interface (show animations, highlight changes)

# Visual Perception

- Vision involves constant movement – of our eyes, head and body, or the things we see or all of the above
- The outer parts of our retina are particularly sensitive to motion, so when something comes into our field of vision, our attention is drawn to it.
- Even the fact that we have two eyes (binocular vision) is a kind of movement: the two views are slightly different, as if we had moved a few inches to the left or right.



# Visual Perception

- Retina is light sensitive and contains 2 types of photoreceptors: rods and cones. Rods are highly sensitive and allow vision of weakly illuminated objects. They dominate peripheral vision
- Cones are grouped in the center and since there are 3 types, sensitive to different wavelength of light they allow color vision

# Visual Perception

- Vision is the main source of information : 75% of information perceived by human is of visual nature
- The fastest signals reach the cortex in about 30 ms. Color perception needs about 80 ms to reach: this is faster than movement perception

# Group Project

- In this assignment, you will start the design your group project by exploring:

## **1. Interface Design and Interaction scenario**

Sketched overall concept of your courseware (windows or dialog boxes, along with menus and controls that the user might need in order to interact with your system).

## **2. System flow**

Use rough sketches to illustrate how the flow of your system would be like.

Keep things simple, focus on the content that you're trying to deliver to the user, what the user needs to do and how they can do it.